Examiner reports

Q1.

Many students scored full marks on (a), though significant numbers scored 1 out of 2 because their conclusion was not drawn clearly enough. In addition, significant numbers failed to score at all because they did not understand the significance of the probabilities in determining independence.

(b) was a very well answered question, with the majority of students scoring full marks, though there was some confusion about the meaning of “mathematics or biology or both”.

Q3.

This fairly standard probability question posed major difficulties to the weaker students whereas the stronger students scored high marks. Almost all students answered part (a)(i) correctly with slightly fewer scoring the mark in part (a)(ii) through the use of multiplication instead of addition. The use of conditional probability to answer parts (a)(iii) and (iv) was beyond many students and it was not unusual to see repeated answers of 0.25 or 0.30. Of those students who scored very few marks in part (a), many were then able to make a sensible and often correct attempt at part (b). Whilst many students identified the need for $(0.30^2 + 0.25^2)$, this was sometimes spoilt by the introduction of 0.55, 0.55^{-1} or 2 as a multiplier.

Q4.

This fairly standard probability question posed major difficulties to the weaker students whereas the stronger students scored high marks often with an apparent minimum of effort. Almost all students answered part (a)(i) correctly with slightly fewer scoring the mark in part (a)(ii) through the use of multiplication instead of addition. The use of conditional probability to answer parts (a)(iii) to (v) was beyond many students and it was not unusual to see repeated answers of 0.25 or 0.30. Of those students who scored very few marks in part (a), many were then able to make a sensible and often correct attempt at part (b). Whilst many students identified the need for $(0.30^2 + 0.25^2)$, this was sometimes spoilt by the introduction of 0.55, 0.55^{-1} or 2 as a multiplier.

Q5.

Answers to this question on probability were, in the main, of a high standard and showed a marked improvement on similar questions in previous papers. It was quite rare to see an incorrect answer to part (a) with $0.90 + 0.95 - 0.855$ in (ii) being by far the most common error. Most students also scored well in part (b) with few students not scoring full marks in (i) to (iii). However, (iv) proved more challenging with $0.10 \times 0.15 + 0.05 \times 0$ as a frequent incorrect answer. Even some better students lost 1 mark due to calculating $0.085 + 0.05$ instead of 0.085×0.05.

Q6.

Almost without exception, full marks were scored in part (a). As a consequence, students usually scored well in part (b) with full marks by far the most common mark. In fact there was no common error but simply one-off methodological or numerical errors. Again in part (c), many students had the correct fractions of $\frac{0.090}{0.155}$ and $\frac{0.065}{0.155}$ or their equivalents.

Where full marks were not scored, it was often due to not including a multiplier of 2.

Q7.

This probability question unexpectedly proved a major or even unachievable challenge to many candidates since they appeared to have no real grasp of a scenario involving two non-mutually exclusive events but instead considered them to be independent events. As a result, common worthless answers to part (a) were: 0.15 × 0.40 = 0.06 ≈ 0.10; (0.85 × 0.40) + (0.15 × 0.60) + 0.55 = 0.98; (0.85 × 0.40) + (0.15 × 0.60) = 0.43. Centres are reminded that the addition law for two non-mutually exclusive events is part of the specification. Those better prepared candidates, who were able to construct a correct 2 × 2 table, draw a correct Venn diagram or use the correct addition law formula, scored at least 2 and quite often all 5 marks. Those candidates using the addition law formula in part (a)(i) sometimes fudged their answer and rarely supplied sufficient detail for the awarding of the full 3 marks. Those candidates scoring few, if any, marks in part (a) faired a little better in part (b) since they were often able to score the 2 marks in part (b)(i) and then one further mark in part (b)(ii) for 0.55 × 0.70. This expression was often accompanied by several incorrect expressions each the product of three probabilities (independence again). The best candidates who scored full or nearly full marks in part (a) usually followed this by a similar impressive performance in part (b), where some very succinct and elegant solutions were seen to part (b)(ii).

Q8.

In answering this question, some students ignored the instruction regarding answers ‘to three decimal places’ and simply left them as fractions. This was not penalised in part (a) but was penalised in part (b). Most students answered parts (a)(i) & (ii) correctly but in part (a)(iii), some incorrectly assumed independence and so multiplied together their previous two answers. Part (a)(iv) caused significantly more students a problem. Whilst many obtained the correct numerator of 153, they used a denominator of 640 instead of 172. Most students scored only 1 or 2 marks in part (b). Whilst some were able to quote the three correct numerators, they then used 194^3 or (640 × 639 × 638) in the denominator or failed to multiply by 3! = 6.

Q9.

In answering this question, a significant proportion of students ignored the instruction regarding answers ‘to three decimal places’ and simply left them as fractions. This was not penalised in part (a) but was penalised in parts (b) & (c). Most students answered parts (a)(i) & (ii) correctly but in part (a)(iii), many incorrectly assumed independence and so multiplied together their previous two answers. Part (a)(iv) caused significantly more students a problem. Whilst many obtained the correct numerator of 153, they used a denominator of 640 instead of 172. Few students were able to make significant progress in part (b). Many responses lacked any attempt at a numerical justification, whilst others tried to identify new values to use from the table rather than following the instruction (and hint) to use answers from part (a). Such answers usually lacked explanation, left answers as fractions that were difficult to compare, or did not consider like-for-like situations (eg B = 3 with $B \le 3$). Most students scored only 1 or 2 marks in part (c). Whilst most were able to quote the three correct numerators, many used 194^3 or (640 × 639 × 638) in the denominator and even more failed to multiply by 3! = 6. Sadly, it was also not rare to see the three correct fractions added together.

Q10.

Answers to this topic continued to show an improvement and it was not unusual to see students scoring at least 12 marks. In part (a), the most common error was to state 0.10 as the answer. It was then rare to find further incorrect answers.

Q11.

Many candidates scored between 5 and 9 marks, with the marks usually being lost in parts (b) and (c). Too many candidates lost valuable marks for ignoring 'to three decimal places'. Fractional answers only scored full marks in parts (a)(i) to (a)(iii) where almost all candidates scored the 3 marks available. In part (a)(iv), the use of P(L ∩ F) or P(F | L), rather than P(L | F), was often in evidence. In part (b), far too many candidates used:

$\left(\frac{94}{126}\right)^2$ or $\frac{94}{126} \times \frac{93}{126}$ instead of $\frac{94}{126} \times \frac{93}{125}$.

In answering part (c), many candidates had the correct numerator of 349 × 193 × 103 but a majority then used a denominator of $(645)^3$ and almost all candidates omitted the permutation multiplier of 3! = 6.

Q12.

Most candidates scored between 7 and 10 marks, with the marks usually being lost in parts (b) and (c). Too many candidates lost valuable marks for ignoring 'to three decimal places'. Fractional answers only scored full marks in parts (a)(i) to (a)(iii), where almost all candidates scored the 3 marks available. In part (a)(iv), P($L \cap F$) or P($F \mid L$), rather than P($L \mid F$), was often found. Similarly, the use of P($L' \cap M$) or P($L' \mid M$) was often in evidence in answers to part (a)(v).

In part (b), far too many candidates used $\left(\frac{94}{126}\right)^2$ or $\frac{94}{126} \times \frac{93}{126}$ instead of $\frac{94}{126} \times \frac{93}{125}$.

In answering part (c), many candidates had the correct numerator of 349 × 193 ×103 × but a majority then used a denominator of $(645)^3$, and almost all candidates omitted the permutation multiplier of 3! = 6.

Q13.

This question on the binomial distribution was a good source of marks for many candidates, with the more able often scoring most of the 15 marks available. Save for a small minority of candidates who calculated P(R = 5) using the formula, part (a)(i) was usually answered correctly from the appropriate table in the supplied booklet. Again in part (a)(ii), answers were usually correct with only a minority of candidates giving P(R ≤ 10), P(R ≤ 9) or 1– P(R ≤ 9) as their answers. Answers to part (a)(iii) were almost always correct with evaluation by formula by far the more common method. As in previous series, part (a)(iv) caused candidates more difficulty, with many uncertain as to how to find P(5 ≤ R ≤ 10). Whilst most candidates did attempt to subtract two cumulative probabilities, they often selected one, or even two, incorrect values. In part (b)(i), most candidates showed correctly that either 0.85 + (0.15 × 0.80) or 1 – (0.15 × 0.20) resulted in 0.97. Incorrect reasoning that also led to the given answer often assumed that that a second shot was always made.

Although many correct answers were seen to part (b)(ii), the logic involved proved too much of a challenge for many candidates. Whilst many identified the correct model of B

(50,0.97), most then attempted P(S = 48) by formula. Most correct answers appeared to be obtained directly from a calculator's inbuilt cumulative binomial function rather than by P(S' ≤ 2) using tables for B(50,0.03). In answering part (b)(iii), most candidates attempted to use np but then failed to identify the correct values of n and p. As a result, 80 × 0.80 = 64 and (80 × 0.15) × 0.80 = 9.6 were common incorrect answers.

Q14.

As expected, it was rare to find an incorrect completion of the table in part (a). Although most candidates identified the correct probabilities of 0.10 and 0.15 in part (b), many then multiplied these two values instead of adding them, or obtained 0.80 from 0.70 + 0.65 – 0.55. Showing justifications of the two statements in part (c) proved too challenging to most candidates, with no attempt or general prose being the most common responses, rather than the necessary numerical justifications. In fact, it was clearly evident from the overall responses that most candidates had little knowledge of, or were confused about, the terms 'mutually exclusive' and 'independent'. Whilst some candidates used P($W \cap J$) = 0.55 to show non–mutually exclusive, very few indeed achieved any marks for justifying non-independence.

Q15.

With the exception of part (a)(iii), candidates' answers scored many, and often all, of the remaining 10 marks. As expected, it was rare indeed to find an incorrect completion of the table in part (a)(i). Although most candidates identified the correct probabilities of 0.10 and 0.15 in part (a)(ii), almost half then multiplied these two values instead of adding them. A minority obtained 0.80 from 0.70 + 0.65 – 0.55 but an even smaller number obtained the correct answer of 0.25 from 0.70 + 0.65 – 2 × 0.55. Showing justifications of the two statements in part (a)(iii) proved too challenging to many candidates, with no attempt or general prose being the most common responses rather than the necessary numerical justifications. In fact, it was clearly evident from the overall responses that most candidates had little knowledge of, or were confused about, the terms 'mutually exclusive', 'exhaustive' and 'independent'. Whilst a minority of candidates used P($W \cap J$) = 0.55 to show non–mutually exclusive, very few indeed achieved any marks for justifying non-independence. Consequently, part (b) proved a welcome relief to most candidates with 4 or 5 marks being by far the most common marks. It was very rare to see an incorrect answer to part (b)(i), but in part (b)(ii) some candidates multiplied 2, instead of 3, probabilities and/or introduced combination multipliers of either 2 or 3.

Q16.

Answers to this probability question were much improved over comparable questions on the same topic in recent papers. Although incorrect answers to part (a)(i) were not unusual
(0.15 × 0.4 + 0.15 × 0.75 + 0.15 × 1 = 0.3225), most candidates scored a large majority of the remaining 6 marks in part (a), with only a small minority failing to consider conditional probability in part (a)(iv). Most candidates then made the correct connection between part (a)(iv) and part (b)(i). It was pleasing to see the significant proportion of candidates who then realised the step for part (b)(ii), with the result that the awarding of full marks in part (b) was not uncommon.

Q17.

Better candidates were able to score at least 9 marks on this question but weaker candidates often failed to score more than 2 or 3 marks. Almost all candidates scored the

2 marks available in part (a)(i). Whilst many candidates also scored the 2 marks in part (a)(ii), a minority over-complicated the request by trying to evaluate 4 terms, although some were eventually successful. For a large proportion of candidates, part (a)(iii) resulted in a loss of 3 of the 4 marks available due to often-correct attempts at P(C = 2) instead of P($C \geq 2$).

Here again some candidates over-complicated the request by trying to evaluate 6 terms rather than 3 for P(C = 2) or 7 terms rather than 4 for P($C \geq 2$). Answers to part (b)(i) were often correct but the simple method for answering part (b)(ii), as 1 – part (b)(i), was sometimes not recognised; the lengthy alternative method often resulted in an incorrect answer.

Q18.

This was another good source of marks for candidates of nearly all abilities. Most candidates correctly multiplied the two probabilities to answer parts (a)(i) and (ii), although 0.03 was regularly seen as the answer to the latter. A common error in part (a)(iii) was to start afresh but to omit the case of both sows and so obtain 0.29 as the answer. Those who recognised the more efficient approach of 1 – (a)(ii) were nearly always correct.

In part (b)(i), most candidates completed the table correctly through simple arithmetic. The usual error was to find P(M') = 0.40 and P(D') = 0.25 correctly through subtraction but then to assume independence by calculating P($M' \cap D'$) as P(M') × P(D'). Despite this making somewhat of a nonsense of the table's row and column totals, the allowing of follow-through answers from tables enabled mostly correct responses to part (b)(ii), although multiplication of two probabilities, rather than their addition, was a common error in the final part.

Q19.

In part (a), most candidates drew a correct tree diagram, even if somewhat untidily, but very few multiplied the probabilities (eg 0.10 × 0.90 = 0.09) which of course should, in total, add to unity; there was no mark penalty for this omission, but some candidates subsequently lost marks through their inaccurate multiplications of decimals. Answers to part (b)(i) were surprisingly weak with, for example, the answer to (A) as 0.02 + 0.0016 = 0.0216.

Follow-through answers to part (b)(ii) often scored full marks, although some candidates either multiplied by 1000, instead of 10 000, or ignored "to the nearest 10". Almost all candidates recognised the need to use Bayes' Theorem in answering part (c). Although there were some fully correct solutions, too many candidates forfeited even follow-through marks through inaccurate arithmetic: surely something that should not occur in a Further Mathematics paper, especially where calculators are permitted.

Q20.

This relatively standard probability question enabled most candidates to score at least 7, sometimes 9 but rarely 11, marks. Full marks were usually scored in parts (a)(i) and (ii). Many candidates also scored full marks in the remainder of part (a) but it was not unusual to see an answer based upon $\frac{60+32}{160}$ for part (a)(ii) and/or answers based upon

$\frac{18}{60}, \frac{48}{160}$ or even $\frac{30}{48}$ for part (a)(iv).

Fully correct answers to part (b) were rare. Although many candidates were able to construct the expression $\frac{24 \times 56 \times 32}{160^3}$ for 1 mark or the expression $\frac{24 \times 56 \times 32}{160 \times 159 \times 158}$ for 2 marks, few realised that permutations had to be considered. Of those who did, about half used 3 rather than 3! = 6. The very small number of candidates who based their correct answer on $\frac{\binom{24}{1}\binom{56}{1}\binom{32}{1}}{\binom{160}{3}}$ are worthy of a special mention!

Q21.

In answering part (a), most candidates gained full or almost full marks. They used a correct expression to deduce the given answer of 0.34 in part (a)(i) and, in part (a)(ii), again correctly used the multiplication law for two dependent events. Similarly, in part (a)(iii), most candidates realised that Bayes' Theorem was required, and so the vast majority scored at least 3 of the 4 marks available.

Part (b) unexpectedly proved to be beyond all but the strongest candidates. The most common solution was to simply evaluate 0.30 × 0.55 × 0.15; candidates should have realised that more was expected for 5 marks. It was expected that part (a)(iii) would lead candidates to evaluate P(City | HB) and P(Country | HB) prior to multiplication. Of the minority who went down this route, very few indeed saw the need to introduce the permutation multiplier of 3! = 6.

Q22.

Answers to part (a) were often fully correct with most candidates capable of applying the general multiplication law and the addition law for mutually exclusive events. Answers to part (b) were much less impressive. Whilst most candidates clearly had some notion of what was needed, their attempts were only partially correct. All too often such answers were the evaluation of $\binom{4}{3}(0.68)^3(0.32)$ or $(0.68)^3(0.105)$.

Q23.

Answers to part (a) were very disappointing and it appeared from candidates' answers that most had no knowledge of probability beyond the application of multiplication laws. However, many candidates scored well in parts (b), (c) and (d), especially as part (c) did require use of the multiplication law (for dependent events). Most candidates scored the mark for part (a)(i) but even here incorrect answers were not a rare event. Answers to parts (a)(ii) & (iii) invariably involved the multiplication law for independent events:

(ii) P (G but not S) = P (G) × P (not S) = 0.315.

(iii) P (G or S only) = (ii) + P (not S) × P(G) = 0.48.

In fact, as realised by some candidates, who were by no means 'high-flyers', the answers required only simple logic as follows:

Since P(S or G or both) = 1 (Andrew uses no other facilities):

(ii) P(G but not S) = 1 – P(S) = 1 – 0.55 = 0.45;

(iii) P(G or S only) = 1 – P(G & S) = 1 – 0.25 = 0.75.

Most candidates realised that the answer to part (b) required $\{(a)(i)\}^4$ and so scored 1 or 2 marks; some had trouble with the number of leading zeros in 0.3^4. A small minority of candidates attempted to consider the number of days in a month. As mentioned above, most candidates applied the correct laws in part (c) and many gained all 3 marks. This was often followed by the correct answer to part (d) although some candidates calculated (1 – 0.35 × 0.45) or (1 – 0.35) × (1 – 0.45).

Q24.

This was a very accessible question for candidates of all abilities. Many candidates scored full marks and the awarding of fewer than 5 marks was rare indeed. Candidates who extracted the necessary information from the table generally had more success than those attempting the use of probability formulae. Few candidates failed to score the 3 marks available for parts (a) and (b). In part (c), common errors were to quote answers of 0.4 + 0.7 = 1.1, 0.4 × 0.7 = 0.28 or 0.4 + 0.7 – 0.28 = 0.82, which appeared to indicate, as on previous papers, a lack of understanding of the addition law for two (non-mutually exclusive and dependent) events; something that needs attention in future. Many candidates coped well with the conditional probabilities in parts (d) and (e). When marks were lost, it was usually for quoting $\frac{42}{400}$ or $\frac{42}{70}$ in part (d) and/or for quoting $\frac{21}{400}$ or $\frac{21}{120}$ in part (e).

Q25.

Most candidates drew adequately a correct tree diagram but very few multiplied the probabilities (eg 0.25 × 0.30) which of course should, in total, add to one; there was no penalty for this 'omission'. However, almost all candidates did the necessary multiplications when answering part (b) correctly and so scored all 4 marks available. Answers to part (c) were much less impressive. Whilst the majority of candidates recognised the need for a binomial model with n = 10, most then used p = {(b)(i)}, 0.15 or 0.55, rather than p = {(b)(ii)}, presumably through not recognising the conditional nature of the request.

Q26.

This question was also quite well answered by a large proportion of candidates. Whilst almost all could answer part (a)(i) correctly, only the more able candidates were capable of dealing with the conditional probabilities as required in parts (a)(ii) and (iii). Thus it was not unusual to see $\frac{6}{50}$ quoted for part (a)(ii) and/or $\frac{25}{50}$ or $\frac{25}{27}$ quoted for part (a)(iii). In answering part (b), it was disappointing to see the number of candidates who apparently ignored the (obvious) implication from the context that the selection had to be '**without**

replacement' but chose instead to quote $\left(\frac{22}{50}\right)^4$ or to employ a binomial distribution!

Q27.

This question was also quite well answered by a large proportion of candidates. Those candidates who extracted the necessary information from the table rather than attempting to apply formulae invariably had the greater success. Whilst almost all could answer parts (a)(i), (ii) & (iii) correctly, only the more able candidates were capable of dealing with the conditional probabilities as required in parts (a)(iv) & (v). Thus it was not unusual to see $\frac{6}{50}$ quoted for part (a)(iv) and/or $\frac{25}{20}$ or $\frac{25}{27}$ quoted for part (a)(v).

In answering part (b), it was disappointing to see the number of candidates who apparently ignored the (obvious) implication from the context that the selection had to be 'without replacement' but chose instead to quote $\left(\frac{22}{50}\right)^4$ or to employ a binomial distribution.

Q28.

A minority of candidates apparently ignored the preliminary explanations and treated the given table as a simple 2-way frequency table, as might be found on an MS1 paper. A significant proportion of the remaining candidates quoted (0.85 × 0.25) as their answer to part (a), apparently ignoring the second paragraph of the question. However, these and many others then proceeded to gain most of, if not all, the marks available for answering parts (b), (c) and (d).

Q29.

This was a very high scoring question with many candidates scoring full marks. Only part (b)(ii) presented any noteworthy difficulty where 0.2 + 0.25 – 0.18 was the most common incorrect approach.

Q30.

Part (a) of this question was, as expected, accessible to all but the weakest candidates. Those candidates who simply extracted the necessary information from the table invariably scored most, if not all, of the 7 marks. Those who used a formula approach were generally less successful through assuming the independence of D and R and/or struggling in the application of a formula to the given information. Answers to part (b)(i) were usually correct but most answers to part (b)(ii) consisted of worthless subjective ramblings. Only the most able candidates attempted to provide quantitative evidence. This often involved showing that P($D \cap R$) was equal to P(D)) x P(R) when a much simpler approach was to use the fact that previous answers to parts (a)(i) and (iii) were equal. Sometimes, despite correct working, candidates then concluded that the two events were not independent. Answers to part (c) were generally of a high standard although, when "not detached" was included, candidates lost 1 mark. A small minority of candidates ignored the phrase "in the context of this question" or evaluated the probabilities of the events and so lost all 4 marks.

Q31.

Part (a) of this question was, as expected, accessible to all but the weakest candidates. Those candidates who simply extracted the necessary information from the table invariably scored most, if not all, of the 9 marks. Those who used a formula approach were generally less successful through assuming the independence of D and R and/or struggling in the application of a formula to the given information. Answers to part (b)(i) were usually correct but most answers to part (b)(ii) consisted of worthless subjective ramblings. Only the most able candidates attempted to provide quantitative evidence. This often involved showing that P(D $\cap$ R) was equal to P(D) x $P(R)$ when a much simpler approach was to use the fact that previous answers to parts (a)(i) and (iv) were equal. Sometimes, despite correct working, candidates then concluded that the two events were not independent. Answers to part (c) were generally of a high standard although, when “not detached” was included, candidates lost 1 mark. A small minority of candidates ignored the phrase “in the context of this question” or evaluated the probabilities of the events and so lost all 4 marks.

Q32.

Overall this was the best answered question on the paper with more than 65% of candidates achieving full marks; a credit to centres’ teaching of Bayes’ Theorem. On the rare occasions when full marks were not scored, it was generally in parts (a)(iii) and/or (b) for using other than Bayes’ Theorem.

Q33.

Somewhat surprisingly perhaps, this was another good source of marks for weaker candidates.

Answers to parts (a)(i), (ii) and (iv) were often correct. Part (a)(iii) was less successfully answered, usually as a result of the duplication of P($F \cap A$) in the calculation.

In part (b)(i), some calculations were based on replacement and so were invalid methods. There was an alarming number of answers greater than 1, candidates having summed their individual probabilities instead of multiplying. However, overall, the success rate was good. Again in part (b)(ii), a few candidates based their calculation on replacement but could receive credit if identifying that there were three possible combinations of MFF. Unfortunately, a sizeable number of candidates thought there was either 1 or 3! such combinations. Otherwise, many candidates were successful.

Answers to part (d) were much less impressive. Although most candidates appeared to have some knowledge of set notation, it was not unusual to see the meanings of $\cap$ and $\cup$ sometimes being swapped or to see $F' \mid A$ interpreted as $A \mid F$. A number of candidates did not answer the questions as set, but merely gave probabilities as their answers or statements such as “F intersected with A”. Candidates must expect to relate, as directed, to the context of the question. Use of “not female” was also not sufficient in this context.

Q34.

This was probably the best answered question on the paper with many candidates, often through first constructing a tree diagram, scoring full marks. It was thus rare to see candidates not scoring the full 7 marks in part (a). When marks were lost, it tended to be in part (a)(ii). Part (b) proved a good discriminator for the more able candidates. Many candidates recognised the need for $p_1^2 \times p_2$ many fewer included the multiplier of 3 and/or had correct values for both p_1 and p_2. A small minority of candidates attempted to apply a binomial model.

Q35.

Many candidates scored full or almost full marks on this probability question, sometimes even those who scored very little elsewhere on the paper. Few candidates failed to score the first 2 marks but, in part (a)(ii), it was not uncommon to see 0.2 × 0.4. In part (b)(i), most candidates scored all 3 marks. However, in part (b)(ii), only 2 of the correct 3 permutations were often used.